Thermobox

A Simple Thermal Control System

Overview

Thermal control systems manage and regulate the temperature within a specific environment or system. They are used in various industries, such as aerospace, automotive, and manufacturing. In this project, a simple thermal control system is modeled, built and demonstrated.

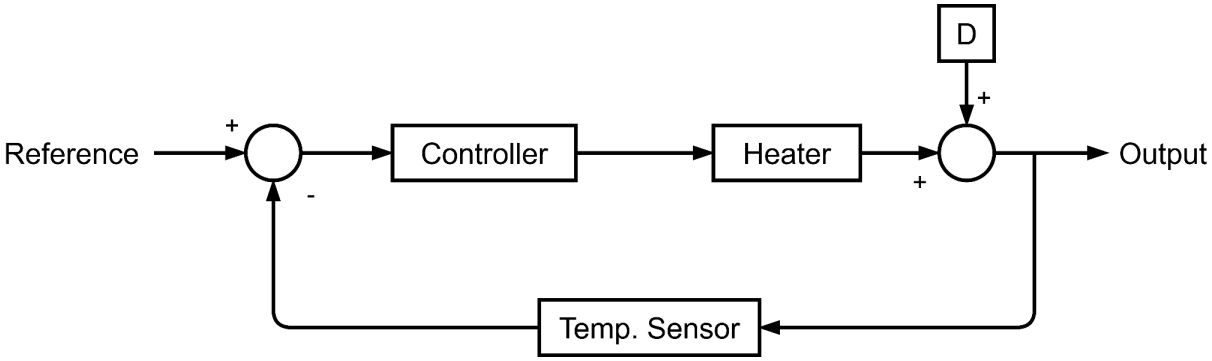


Figure: General block diagram of the system.

Goal & Deliverables

The goal of this project is to minimize settling time and overshoot while rejecting disturbances. Model parameters will be estimated and PI control will be utilized. In order to achieve this goal, multiple project deliverables have been defined.

Table: Project deliverables.

#	Deliverable	Projected Deadline
1	Simulation of system in Simulink	23 Sept 2026
2	Performance & stability evaluation of system using frequency response theory	30 Sept 2026
3	System diagram with every component	30 Sept 2026
4	Block diagram representation of system	4 Oct 2026
5	Final system operation (in lab)	9 Oct 2026
6	Proper documentation with GitHub repo	23 Oct 2026
7	System demonstration video	24 Oct 2026
8	Practical report (written in LaTeX)	31 Oct 2026

System Blocks

Input & Output

The reference input is variable (a setpoint temperature) and the output of this system is temperature, which is sensed and compared to the reference, thereby creating an error.

PI Controller

PI means Proportional and Integral and is the controller type that will be utilized for this project. For the simulation, a built-in PID controller block can be used with $K_D = 0$. For the physical version, a program is necessary to logically mimic the PI functionality.

Heater

The heater consists of a power resistor and a circuit to drive it. The figure below illustrates the driver circuit. The NPN transistor is necessary to amplify the control signal (from the Arduino), allowing for full MOSFET saturation.

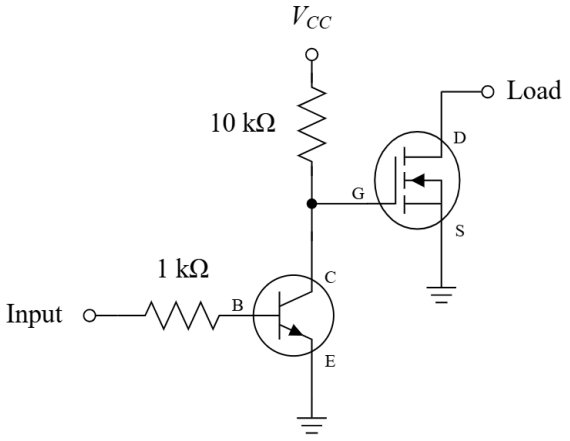


Figure: Heater driver circuit.

Disturbance

The disturbance occurs after the plant as it affects the temperature within the box. In this case, the foam box can be temporarily opened. However, for the simulation, a secondary step input (the first is the reference) will be used.

Project Requirements

Table: Required components.

#	Component	Qty.	Specifications	Application
1	Arduino Uno	1	- 6 AIOs - 14 DIOs (6 w/ PWM) - V_{CC} = 5 V (via USB) - V_{CC} = 6 - 20 V (via V_{IN})	Programmable microcontroller containing the control loop logic.
2	DS18B20 temperature probe	1	- ± 0.5 °C accuracy - Operational at V_{CC} = 5 V	Measures the output of the system (temperature) and is located in the feedback loop.
3	YAGEO Through Hole Power Resistor	1	- 6.2 Ω ± 5% - 10W - c_p = 800 J/(kg K) - m ≈ 8 g	Functions as the plant (heater) connected to 7.5 V.
4	IRF44ZN MOSFET	1	- Fast Switching - V_{GS} ≈ 10 V (For full sat.)	Part of the plant. Functions as a driver for the power resistor as it pulls a lot of current.
5	NPN 2N3904 Transistor	1	- β (h_{fe}) = 100 - 400 - Max. V_{CB} = 60 V - Max. V_{BE} = 6 V - Max. I_C = 200 mA	Amplifies signal going into the MOSFET.
6	Resistors	3	- 1× 1 kΩ - 1× 4.7 kΩ - 1× 10 kΩ	Pull-up, pull-down or current-limiting resistors.
7	Wires & Jumper cables	TBA	- R ≈ 0.1 - 1 Ω (negligible) - AWG 24 - I_{MAX} = 3.5 A	Connections between the various system elements.
8	Breadboard	1	- I_{MAX} = 1 A - I_{TYP} = <500 mA	For the heater driver circuit.
9	Small foam box	1	- κ = 0.033 W/(m K) - $V_{meas.}$ ≈ 0.767 L	Isolated space from the outside.

Table: Required equipment.

Instrument			Qty.	Specifications	Application
#	Brand	Model			
1	Siglent	SPD3303C	1	- Max. V_{OUT} = 32 V - Max. I_{OUT} = 3.2 A	DC power supply
2	Fluke	175 True RMS Multimeter	1	- ± 0.15% accuracy (6 - 600 V range) - ± 0.9% accuracy (600 Ω - 60 kΩ range)	To check resistor values and voltages.
3	—	Measurement clamps	2	—	Used with DMM to measure accurately.